Wavefunction Collapse as Retrodiction Failure: Gravitational Decoherence, Objectivity, and the τ-R Bridge

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We apply the temporal asymmetry parameter $\tau = 1 - F$ from the Petz recovery framework [S.-K. Huang, companion paper] to the gravitational dephasing channel of Pikovski et al., obtaining an explicit, time-dependent quantification of "collapse" as Petz recovery failure. For a system with N internal degrees of freedom, $\tau(t) = 1 - \exp(-N\Gamma t^2)$ and the quantum Darwinism redundancy $R(t) = N[1 - \exp(-\Gamma t^2)]$ are governed by the same parameter $N\Gamma$, establishing a τ-R *bridge*: irreversibility ($\tau \to 1$) and objectivity ($R \to N$) emerge at the same timescale. Under the Diósi–Penrose assumption, the collapse timescale reduces to $t_c = \hbar/E_G$. We connect $\tau = 1$ to operational superselection via the Englert–Greenberger wave-particle duality relation and spectrum broadcast structure, providing a quantitative language for collapse within standard quantum mechanics, with explicit bounds on spatial recoverability. Predictions for molecule interferometry, optomechanical, and gravitational entanglement experiments are tabulated and shown to be consistent with all existing data. The universal bound $F_{\text{spatial}}(t) \geq \exp(-\Sigma_{\text{grav}}(t)/2)$ constrains the recoverability of spatial coherence for any gravitational channel.

INTRODUCTION

In a companion paper [1], we established that the Petz recovery map [2, 3] unifies quantum erasure, Bayesian retrodiction, quantum error correction, and thermodynamic irreversibility. The temporal asymmetry parameter

$$\tau = 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big) \tag{1}$$

quantifies the degree to which a quantum channel $\mathcal{N}$ distinguishes past from future: $\tau = 0$ for closed (unitary) evolution, $\tau > 0$ for open systems, with the bound $\tau \leq 1 - \exp(-\Sigma/2)$ linking recovery failure to entropy production Σ [4, 5].

A central question in quantum foundations remains: when does a quantum superposition become a classical mixture? The decoherence program [6–8] identifies environment-induced decoherence as the mechanism, but leaves open what distinguishes "for all practical purposes" (FAPP) collapse from genuine irreversibility. Meanwhile, Penrose [9] and Diósi [10] have argued that gravitational self-energy sets a fundamental collapse timescale $t_c = \hbar/E_G$.

In this Letter, we show that the τ framework provides a precise, quantitative answer. We define *operational collapse* as $\tau \to 1$: the Petz recovery map—the unique Bayesian retrodiction functor [12]—fails completely, certifying that no physical protocol can recover the prior superposition. We derive $\tau(t)$ explicitly for the gravitational dephasing channel of Pikovski et al. [11] [Sec.], connect internal and spatial decoherence via an entanglement complementarity bridge [Sec.], establish a quantitative link between irreversibility and objectivity (the τ-R bridge) [Sec.], recover the Penrose timescale under the Diósi–Penrose assumption [Sec.], and tabulate predictions for current and proposed experiments [Sec.]. Each result is labeled [KNOWN], [NEW SYNTHESIS], or [NEW FORMULA].

GRAVITATIONAL DEPHASING AS A QUANTUM CHANNEL

Pikovski Hamiltonian [KNOWN].—Consider a massive particle in a spatial superposition $|\Psi(0)\rangle = (\alpha|x_1\rangle + \beta|x_2\rangle) \otimes |\chi_{\text{int}}\rangle$, where $|\chi_{\text{int}}\rangle$ is the internal state. In the post-Newtonian approximation, the total Hamiltonian is [11]

$$H = \frac{p^2}{2m} + H_{\text{int}}\left(1 + \frac{\Phi(\hat{x})}{c^2}\right), \tag{2}$$

where $\Phi(\hat{x})$ is the Newtonian gravitational potential evaluated at the center-of-mass position. Time evolution under Eq. (2) produces [11]

$$|\Psi(t)\rangle = \alpha|x_1\rangle \otimes U_1|\chi\rangle + \beta|x_2\rangle \otimes U_2|\chi\rangle, \tag{3}$$

where $U_j = \exp(-iH_{\text{int}}(1+\Phi_j/c^2)\,t/\hbar)$ with $\Phi_j \equiv \Phi(x_j)$.

Decoherence factor [KNOWN].—Tracing over internal degrees of freedom yields the reduced spatial state

$$\rho_S(t) = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* D(t) \\ \alpha^*\beta\, D^*(t) & |\beta|^2 \end{pmatrix}, \tag{4}$$

with decoherence factor

$$D(t) = \langle\chi|\exp\left(\frac{iH_{\text{int}}\Delta\Phi\, t}{\hbar c^2}\right)|\chi\rangle = \sum_n |c_n|^2 e^{-iE_n\Delta\Phi\, t/(\hbar c^2)}, \tag{5}$$

where $\Delta\Phi = \Phi(x_1) - \Phi(x_2)$, $\{E_n\}$ are internal energy eigenvalues, and $c_n = \langle E_n|\chi\rangle$.

CPTP dephasing channel [KNOWN].—The map $\mathcal{N}_{\text{grav}} : \rho_S(0) \mapsto \rho_S(t)$ is a completely positive, trace-preserving

(CPTP) dephasing channel with Kraus operators [8]

$$K_0 = \sqrt{\frac{1+p}{2}}\, I, \qquad K_1 = \sqrt{\frac{1-p}{2}}\, \sigma_z, \tag{6}$$

where $p(t) = |D(t)|$ and $\sigma_z = |x_1\rangle\langle x_1| - |x_2\rangle\langle x_2|$. One verifies $K_0^\dagger K_0 + K_1^\dagger K_1 = I$ and $\mathcal{N}_{\rm grav}(\rho)_{12} = p(t)\,\rho_{12}$.

Many internal modes [KNOWN].—For N internal modes in a thermal state at temperature T, the decoherence factor factorizes: $D(t) = \prod_{k=1}^N d_k(t)$, where each mode contributes $d_k(t) = 1 - \varepsilon_k(t)$ with $\varepsilon_k \ll 1$. The product yields

$$|D(t)| \approx \exp\left(-\sum_k \varepsilon_k\right) = \exp(-N\langle\varepsilon\rangle), \tag{7}$$

giving Gaussian decay $p(t) = \exp(-\Gamma t^2)$ with $\Gamma = (\Delta E)^2(\Delta\Phi)^2/(2\hbar^2 c^4)$ for a thermal internal state with energy variance $(\Delta E)^2$ [11].

Temporal asymmetry parameter [NEW SYNTHESIS].—The Petz recovery map for the dephasing channel with maximally mixed reference state is self-adjoint: $\mathcal{R} = \mathcal{N}_{\rm grav}$. Hence the recovery fidelity (worst case, $|+\rangle$ input) is $F_{\rm Petz} = (1+p^2)/2$. Using the universal (rotated) Petz map [4] and defining $\tau(t) = 1 - F(t)$, we obtain

$$\boxed{\tau_{\rm grav}(t) = 1 - \exp\big(-N\Gamma t^2\big)} \tag{8}$$

for N internal modes in the large-N limit. This is a monotone from $\tau = 0$ (full coherence) to $\tau \to 1$ (operational collapse), with characteristic timescale

$$t_{\rm dec} = \frac{1}{\sqrt{N\Gamma}} = \frac{\hbar c^2\sqrt{2}}{\sqrt{N}\,\Delta E\,|\Delta\Phi|}. \tag{9}$$

Entropy production [KNOWN].—The entropy production of the gravitational dephasing channel is $\Sigma(t) = H_{\rm bin}((1+p(t))/2)$, where $H_{\rm bin}(x) = -x\log x - (1-x)\log(1-x)$ is the binary entropy. The bound $\tau \le 1 - \exp(-\Sigma/2)$ from Ref. [1] is satisfied identically.

INTERNAL–SPATIAL BRIDGE

[NEW SYNTHESIS] The state (3) is a pure bipartite state on $\mathcal{H}_{\rm spatial} \otimes \mathcal{H}_{\rm internal}$. Partial traces define two dephasing channels: $\mathcal{N}_S = {\rm Tr}_{\rm int}$ (yielding ρ_S) and $\mathcal{N}_I = {\rm Tr}_{\rm spatial}$ (yielding ρ_I). Define τ_S and τ_I as the temporal asymmetry parameters for each channel, evaluated for the worst-case input (equal superposition).

For the spatial channel, $\tau_S = (1 - |D(t)|)/2$. For the internal channel, $\tau_I = (1 - |D(t)|^2)/2$. Eliminating $|D|$ yields the *bridge relation*:

$$\boxed{\tau_I = 2\,\tau_S\,(1-\tau_S).} \tag{10}$$

This is a concave parabola with maximum $\tau_I = 1/2$ at $\tau_S = 1/2$, reflecting the fact that a qubit dephasing channel saturates at $\tau = 1/2$.

Entanglement entropy [KNOWN].—Since the total state is pure, the von Neumann entropy of either subsystem equals the entanglement entropy:

$$S_{\rm ent}(t) = H_{\rm bin}\big(1 - \tau_S(t)\big). \tag{11}$$

The entropy production of both channels equals the entanglement entropy:

$$\Sigma_{\rm int} = \Sigma_{\rm spatial} = S_{\rm ent}. \tag{12}$$

This identity [NEW SYNTHESIS] follows because both channels arise from tracing out the complementary subsystem of a pure state, so the DPI decrease ΔD equals $S_{\rm ent}$ in both cases.

Englert–Greenberger complementarity [NEW SYNTHESIS].—The wave-particle duality relation of Englert and Greenberger [16, 17] takes the form $\mathcal{V}^2 + \mathcal{D}_{\rm wp}^2 \le 1$, where $\mathcal{V}$ is visibility and $\mathcal{D}_{\rm wp}$ is which-path distinguishability. In the τ language:

$$\mathcal{V} = 1 - 2\tau_S, \qquad \mathcal{D}_{\rm wp} = \sqrt{2\tau_I}. \tag{13}$$

The duality relation becomes $\mathcal{V}^2 + \mathcal{D}_{\rm wp}^2 = (1-2\tau_S)^2 + 2\tau_I = 1$, which follows identically from Eq. (10). When $\tau_I = 1/2$ (the internal degrees of freedom carry complete which-path information), $\mathcal{V} = 0$: spatial coherence vanishes. This is operationally indistinguishable from "collapse" into a position eigenstate.

Pointer basis from the equivalence principle [NEW SYNTHESIS].—The equivalence principle dictates that the gravitational potential $\Phi(\hat{x})$ in Eq. (2) couples to *position*: the internal Hamiltonian acquires a position-dependent correction $H_{\rm int}\cdot\Phi(\hat{x})/c^2$. By Zurek's einselection criterion [6], the interaction Hamiltonian selects the pointer basis as the eigenbasis of the system observable that commutes with the coupling. Since $\Phi(\hat{x})$ is diagonal in position, gravity universally selects *position* as the pointer basis. Unlike electromagnetic or nuclear decoherence, where the pointer basis depends on the details of the specific interaction, gravitational einselection is universal—it applies to any massive system regardless of its internal composition. This explains why macroscopic objects always have definite positions (not definite momenta or energy superpositions): the equivalence principle singles out position as the only gravitationally robust observable.

FROM DECOHERENCE TO OBJECTIVITY: THE τ-R BRIDGE

The previous sections established that $\tau \to 1$ certifies irreversibility. We now show that $\tau \to 1$ and objectivity

(in the sense of quantum Darwinism [13]) emerge from the *same* physical parameter.

Tensor product amplification [KNOWN].—The decoherence factor (7) is a product over N independent modes. Each mode records which-path information $\varepsilon_k \ll 1$, but the product decays exponentially: $|D| \approx \exp(-N\langle\varepsilon\rangle)$. This exponential-in-$N$ suppression is a direct consequence of the tensor product structure of quantum mechanics (an axiom, not derived from dynamics) [7, 8]. Even for $\langle\varepsilon\rangle \sim 10^{-20}$, the factor $N \sim 10^{23}$ of a macroscopic object drives $|D| \to 0$ and $\tau \to 1$.

Quantum Darwinism redundancy [KNOWN].—In the framework of quantum Darwinism [13], the redundancy R counts the number of independent environmental fragments that each contain sufficient information to determine the system's state. For the gravitational dephasing channel with N internal modes, each mode acquires which-path information at rate $\Gamma_{\text{single}} = \Gamma/N$. A fragment of size $f \cdot N$ modes gives partial decoherence factor $|D_f| = \exp(-fN\Gamma_{\text{single}}t^2) = \exp(-f\Gamma t^2)$, which becomes negligible when $f > f_c = 1/(\Gamma t^2)$. The redundancy is [13, 14]

$$R(t) = \frac{1}{f_c} = N\big[1 - \exp(-\Gamma_{\text{single}}t^2)\big]\,. \tag{14}$$

The τ-R bridge [NEW FORMULA].—Comparing Eqs. (8) and (14), both are governed by the same dephasing parameter Γ_{single}:

$$\boxed{\begin{aligned} \tau(t) &= 1 - \exp\big(-N\Gamma_{\text{single}}t^2\big), \\ R(t) &= N\big[1 - \exp(-\Gamma_{\text{single}}t^2)\big]\,. \end{aligned}} \tag{15}$$

At the decoherence time $t_{\text{dec}} = 1/\sqrt{N\Gamma_{\text{single}}}$, we have $\tau(t_{\text{dec}}) = 1 - 1/e \approx 0.63$ and $R(t_{\text{dec}}) \approx N(1 - e^{-1/N}) \approx 1$ for $N \gg 1$. At time $t = \sqrt{\ln N/\Gamma_{\text{single}}}$, $\tau \to 1 - N^{-N} \approx 1$ and $R \to N(1 - 1/N) \approx N$. This demonstrates that *collapse ($\tau \to 1$) and objectivity ($R \to N$) occur at the same timescale*—they are two manifestations of the same exponential-in-N decoherence process.

Spectrum broadcast structure (SBS) [NEW SYNTHESIS].—Korbicz et al. [15] proved that any von Neumann-type interaction with $N \gg 1$ sub-environments generically produces spectrum broadcast structure:

$$\rho_{SE}(t) = \sum_i p_i\, |i\rangle\langle i|_S \otimes \rho_i^{(1)} \otimes \rho_i^{(2)} \otimes \cdots \otimes \rho_i^{(N)}, \tag{16}$$

where $\{|i\rangle\}$ are the pointer states and $\rho_i^{(k)}$ are distinguishable states for each environmental fragment. The gravitational dephasing interaction (2) is a von Neumann measurement of position by internal degrees of freedom: $H_{\text{int}} = \sum_k E_k|E_k\rangle\langle E_k|$ coupled to $\Phi(\hat{x})$. By the Korbicz theorem, this produces SBS with pointer states $\{|x_1\rangle, |x_2\rangle\}$. SBS is the strongest known condition for "objective existence" [15]: it ensures (i) the system is in a definite pointer state, (ii) multiple observers independently agree on the outcome, and (iii) observation does not disturb the state.

Why gravity is special [NEW SYNTHESIS].—Among all interactions that produce decoherence, gravity occupies a unique position. By the equivalence principle, gravity couples universally to all forms of energy-momentum: no form of matter or radiation is gravitationally neutral [9]. Furthermore, gravity cannot be shielded—there is no gravitational analogue of a Faraday cage [11]. Every other source of decoherence (electromagnetic, nuclear, thermal) can, in principle, be eliminated by sufficient isolation. Gravitational dephasing cannot. Consequently, τ_{grav} constitutes the *irreducible floor* of temporal asymmetry: for any massive system in a spatial superposition, $\tau_{\text{total}} \geq \tau_{\text{grav}} > 0$. This is the fundamental lower bound on collapse for massive objects, independent of engineering improvements in isolation. The universality of gravity (coupling to all energy-momentum) together with its unshieldability makes gravitational decoherence qualitatively different from all other decoherence mechanisms—it is the only source that is both universal and inescapable [9, 11, 19].

$\tau = 1$ and operational superselection [NEW SYNTHESIS].—When $\tau = 1$, no recovery map $\mathcal{R}$ exists such that $\mathcal{R} \circ \mathcal{N} \approx \text{id}$. This means no observable in the accessible algebra (spatial observables after tracing out internal degrees of freedom) can distinguish ρ_S from the decohered state $\rho_S^{\text{dec}} = |\alpha|^2|x_1\rangle\langle x_1| + |\beta|^2|x_2\rangle\langle x_2|$. This is operationally equivalent to a *superselection rule* on position [18]: the coherent superposition is operationally meaningless, and the system behaves as if in one definite position.

For gravitational decoherence specifically, the connection to superselection is strengthened by two results: (i) Danielson, Satishchandran, and Wald [19] showed that superpositions of masses radiate soft gravitons that cross the cosmological horizon—information is causally irretrievable, not merely practically inaccessible. (ii) Kiefer and Joos [20] proved that with infinitely many graviton field modes, the decoherence is permanent (no Poincaré recurrence). Together with the infrared superselection structure [21], $\tau = 1$ from gravitational dephasing may be exact rather than approximate, providing genuine (not FAPP) irreversibility within standard quantum mechanics.

CONNECTION TO THE PENROSE TIMESCALE

The Diósi–Penrose assumption [KNOWN].—Diósi and Penrose independently proposed that the gravitational

self-energy of a spatial superposition,

$$E_G = G \int d^3\mathbf{x}\, d^3\mathbf{x}' \, \frac{[\rho_1(\mathbf{x}) - \rho_2(\mathbf{x})][\rho_1(\mathbf{x}') - \rho_2(\mathbf{x}')]}{|\mathbf{x} - \mathbf{x}'|}, \tag{17}$$

where ρ_1, ρ_2 are mass distributions corresponding to $|x_1\rangle, |x_2\rangle$, sets a fundamental collapse rate $\Lambda = E_G/\hbar$ [9, 10].

Penrose timescale from τ [NEW SYNTHESIS].—Under the Diósi–Penrose assumption, the decoherence factor takes exponential form: $|D(t)| = \exp(-E_G t/\hbar)$. Substituting into Eq. (8) (with the identification $N\Gamma t^2 \to E_G t/\hbar$ for the Diósi–Penrose mechanism), the collapse criterion $\tau(t_c) = 1 - 1/e$ gives

$$t_c = \frac{\hbar}{E_G}\,, \tag{18}$$

reproducing the Penrose timescale. For a sphere of mass m and radius R displaced by $\Delta x \gg R$, $E_G \approx Gm^2/R$, giving $t_c \approx \hbar R/(Gm^2)$. For a 10^{-14} kg object ($R \sim 10\ \mu$m), $t_c \sim 15$ s; for a 2×10^{-4} kg mirror ($R \sim 1$ mm), $t_c \sim 1.5\ \mu$s [9].

Honest limitations [KNOWN].—Three caveats are essential:

(i) The Pikovski mechanism [11] and the Diósi–Penrose mechanism [9, 10] are *different physical processes*. Pikovski dephasing arises from gravitational time dilation coupling to internal degrees of freedom; the Diósi–Penrose mechanism invokes the gravitational self-energy of the superposition itself. They coincide only when the internal energy variance satisfies $\Delta E \cdot |\Delta\Phi|/(\sqrt{2}\,c^2) = E_G$. Our τ framework accommodates both as inputs but does not derive E_G from first principles.

(ii) E_G is an input parameter, not an output. The τ framework provides the operational definition of collapse ($\tau \to 1$) and the thermodynamic bound ($F \geq \exp(-\Sigma/2)$), but it does not predict the decoherence rate; it takes the rate as given and quantifies its consequences.

(iii) Galley, Giacomini, and Selby [22] proved that any consistent coupling between classical gravity and quantum matter is *fundamentally irreversible*—no Stinespring dilation exists. If gravity has classical aspects, then $\tau > 0$ from gravitational dephasing is genuine (not FAPP), because the channel has no unitary extension. If gravity is fully quantum, the total system admits a unitary dilation, and $\tau > 0$ is FAPP—but operationally indistinguishable in all current experiments.

EXPERIMENTAL PREDICTIONS

[NEW SYNTHESIS] Table I collects the gravitational temporal asymmetry $\tau_{\rm grav}$ for major experiments. In every case, the observed decoherence rates greatly exceed the gravitational floor, consistent with $\tau_{\rm grav}$ being a lower bound on total temporal asymmetry.

TABLE I. Gravitational temporal asymmetry $\tau_{\rm grav}$ for current and proposed experiments. $t_{\rm DP} = \hbar/E_G$ is the Diósi–Penrose timescale. $t_{\rm exp}$ is the experimental coherence time. All current experiments have $\tau_{\rm grav} \ll 1$.

Experiment	m (kg)	$t_{\rm DP}$ (s)	$\tau_{\rm grav}$
Fein 2019 [23]	4×10^{-23}	$\sim 10^{14}$	$\ll 10^{-14}$
Arndt 2025 [24]	3×10^{-22}	$\sim 10^{12}$	$\ll 10^{-12}$
Delić 2020 [25]	3×10^{-18}	$\sim 10^{5}$	$\ll 10^{-5}$
Panda 2024 [26]	2×10^{-25}	$\sim 10^{21}$	$\ll 10^{-19}$
BMV (proposed) [27]	10^{-14}	~ 15	~ 0.004
Tagg mirror [9]	2×10^{-4}	$\sim 1.5\ \mu$s	~ 1

Universal bound on spatial recoverability [NEW SYNTHESIS].—From the master inequality chain of Ref. [1], any gravitational dephasing channel satisfies

$$F_{\rm spatial}(t) \geq \exp\big(-\Sigma_{\rm grav}(t)/2\big). \tag{19}$$

This constrains the performance of *any* recovery protocol (including hypothetical ones beyond current technology): spatial coherence cannot be restored beyond the limit set by gravitational entropy production.

τ-R bridge prediction [NEW FORMULA].—The bridge (15) predicts that collapse ($\tau \to 1$) and objectivity ($R \to N$) emerge at the same timescale. This is testable via quantum Darwinism experiments [30, 31]: by monitoring both the decoherence of the system and the buildup of redundant information in environmental fragments, one can verify that τ and R track the same underlying parameter $N\Gamma_{\rm single}$.

BMV experiment [KNOWN].—The Bose–Marletto–Vedral experiment [27, 28] proposes to detect gravitationally-induced entanglement between two masses. In our framework, this corresponds to the regime $\tau_{\rm grav} \sim 0.004$ (for $t = 1$ s, $m = 10^{-14}$ kg), where partial decoherence is measurable but collapse is far from complete. A positive result (entanglement detected) would demonstrate that $\tau < 1$ for the gravitational channel and that gravity mediates quantum correlations. A negative result would be consistent with $\tau > 0$ being genuine (Galley–Giacomini–Selby [22]).

Consistency with existing data.—All current experiments satisfy $\tau_{\rm grav} \ll 1$, and observed decoherence rates are orders of magnitude larger than the gravitational prediction. The parameter-free Diósi model is ruled out by underground experiments [29], but Penrose's version (with nuclear-scale smearing) remains viable. No experiment contradicts the bound (19).

DISCUSSION

We have shown that wavefunction collapse—the transition from quantum superposition to classical definiteness—admits a precise information-theoretic characterization: *operational collapse is Petz recovery failure* ($\tau \to 1$). The Petz recovery map, being the unique Bayesian retrodiction functor [12], certifies that no physical protocol can infer the prior superposition from the decohered state.

What we answer.—Three questions receive quantitative answers within this framework:

(i) *Why does asymmetry cause collapse?* The tensor product structure of quantum mechanics amplifies any $\varepsilon_k > 0$ per environmental mode into exponential-in-N suppression of coherence [7]. This is not an approximation but a mathematical consequence of the Hilbert space axiom.

(ii) *Why operationally only one outcome?* When $\tau = 1$, no observable in the accessible algebra distinguishes the state from a classical mixture. Combined with spectrum broadcast structure (SBS) [15], this ensures that all observers independently agree on one outcome—the operational definition of objectivity.

(iii) *What is the timescale?* The τ-R bridge (15) shows that collapse and objectivity emerge at $t_{\rm dec} = 1/\sqrt{N\Gamma_{\rm single}}$. Under the Diósi–Penrose assumption, this reduces to the Penrose timescale $t_c = \hbar/E_G$.

What we do not answer.—The *ontological* question (whether other branches "cease to exist") lies outside the framework. Our definition of collapse is operational: it concerns what can be physically measured and recovered, not what "truly exists." The distinction between operational and ontological collapse is a choice of interpretation (Everett, Copenhagen, etc.), not a question that τ can settle. We note, however, that the operational answer is the strongest statement physics can make without additional metaphysical commitments.

Open questions and limitations.—We identify five limitations of the present framework that deserve explicit acknowledgment.

(i) *Born rule.* The framework assumes the Born rule in computing both F (fidelity) and Σ (entropy production). A complete theory of collapse should derive the Born rule from more primitive axioms. We do not attempt this; our results are conditional on standard quantum probability.

(ii) *Intermediate regime.* The physics of $0 < \tau < 1$ (partial collapse) deserves further investigation. In this regime the system is neither fully quantum ($\tau = 0$) nor fully classical ($\tau = 1$). What operational statements can be made about a system with, say, $\tau = 0.3$? The bound $F \geq \exp(-\Sigma/2)$ constrains recovery, but a richer operational characterization of intermediate τ remains an open problem.

(iii) *Observer dependence.* Different observers—defined by different system-environment partitions—assign different τ to the same process. In a Wigner's friend scenario, Wigner assigns $\tau = 0$ to a process for which Wigner's friend assigns $\tau = 1$. The framework is internally consistent (τ is partition-dependent by construction), but the relationship between partition-dependent τ values and intersubjective agreement requires further analysis.

(iv) *Separating gravitational decoherence.* In all current experiments, non-gravitational decoherence dominates by many orders of magnitude (Table I). Isolating the gravitational contribution $\tau_{\rm grav}$ requires either extremely massive quantum superpositions (e.g., the BMV proposal [27]) or space-based experiments with extraordinary isolation from electromagnetic and thermal noise. Until $\tau_{\rm grav}$ is directly measured, the gravitational predictions of this framework remain untested.

(v) *Scope of claims.* This framework provides a quantitative language for collapse (τ) and a universal bound ($F \geq \exp(-\Sigma/2)$), not a resolution of the measurement problem. The measurement problem asks "why one outcome?"; we answer "when recovery becomes impossible." These are related but distinct questions: the former is ontological, the latter operational.

Relationship to Paper 1.—This work specializes the general τ framework of Ref. [1] to gravitational channels, demonstrating that the quantum eraser–retrodiction–QEC–thermodynamics unification extends naturally to the collapse problem. The equivalence chain $\tau = 0 \Leftrightarrow \Sigma = 0 \Leftrightarrow I(A;E|B) = 0 \Leftrightarrow$ QMC [1] applies directly: gravitational decoherence breaks all four conditions simultaneously.

Outlook.—The τ-R bridge suggests a broader program. In a companion paper [32], we apply the same framework to strong gravitational fields, where $\tau = r_s/r$ and the Petz recovery bound constrains the structure of spacetime near horizons. The connection between decoherence and objectivity established here may also inform quantum error correction strategies: quantum error correction can be understood as the deliberate construction of a local environment where $\tau \approx 0$—an artificial suppression of the arrow of time.

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